CSIT127 Networks and Communications

Group Report

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<u>Group Contribution Chart</u>

Name	Contributed Content
Megan Michaelis Leqi Phua	**1.** Introduction **2.** Requirements Analysis **3.** Topology (Explanations Only) **6.** Network Devices & Connectivity **7.** Hardware & Equipment Costing **8.** Server Design **9.1** Staff and Student Access Control **11.** Budget Breakdown **12.** Conclusion Final report editing and compilation
P Reshmah	**9.** Security Policies **10.** Facilities Layout Final report editing and compilation
Wang Wenjie	**3.** Network Topology Design & Solution **7.** Hardware & Equipment Costs **11.** Budget Breakdown Final report editing and compilation
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1. Introduction

1.1. Project Background

A new education campus is planned to accommodate approximately 300 students across two schools: School A and School B. School A has 100 students and 10 teaching staff, while School B has 200 students and 10 teaching staff. In addition to the two schools, the campus also includes supporting departments such as Human Resources (HR), Finance, and IT Support, with 10 staff members in each department (CSIT127 Assignment Brief, 2026).

As digital learning and online communication become increasingly important in educational environments, a reliable and secure campus network infrastructure is required to support teaching, learning, and administrative activities. The proposed network must provide stable Internet connectivity, communication services, email access, file sharing, and secure access to internal resources for both students and staff.

This project focuses on designing a scalable and cost-effective campus network infrastructure that uses appropriate network topologies, VLAN segmentation, IPv4 subnetting, servers, routers, switches, and security mechanisms. The proposed solution aims to improve communication efficiency, network security, and overall manageability while remaining within the allocated budget of SGD 800,000.

1.2. Objectives

The objectives of this project are:

- To design a secure and scalable campus network infrastructure for School A, School B, and supporting departments.
- To implement appropriate network topologies to enable efficient communication among classrooms, laboratories, offices, and departments.
- To perform IPv4 addressing and subnetting using the 10.0.0.0/8 network range.
- To implement VLAN segmentation to separate students, staff, departments, and servers for improved security and traffic management.
- To propose suitable networking devices, such as routers, switches, and servers, as well as Ethernet cabling, for the campus environment.
- To provide Internet access, email services, file sharing, and server connectivity for staff and students.
- To implement basic network security measures, including access control and department isolation.
- To ensure the proposed network design remains within the allocated project budget while supporting future scalability and expansion.

2. Requirement Analysis

2.1. School A

School A consists of approximately 100 students and 10 teaching staff members. The network infrastructure for School A must support daily academic activities such as online learning, Internet access, email communication, file sharing, and access to campus resources.

The network should support:

- Desktop PCs and laptops
- Wireless connectivity through Wi-Fi access points
- Smart classroom devices and projectors
- Shared printers and photocopying machines
- IP phones for staff communication

- Access to web, file, DNS, DHCP, and email servers

To improve security and network management, VLAN segmentation will be implemented to separate student and staff networks.

2.2. School B

School B has 200 students and 10 teaching staff members, making it the larger academic division on campus. Due to the higher number of users, the network must provide stable high-speed connectivity and sufficient bandwidth to support simultaneous access to online learning platforms and campus services.

The network should support:

- Computer laboratories and classroom systems
- High-speed wired and wireless Internet access
- Smart displays and multimedia learning devices
- Shared network printers
- IP telephony services
- Access to centralised servers and cloud-based learning resources

VLAN segmentation and subnetting will be implemented to improve traffic management, scalability, and security between students, staff, and other departments.

2.3. Supporting Departments

The campus includes supporting departments such as Human Resources (HR), Finance, and IT Support, with approximately 10 staff members in each department. These departments require secure and reliable network connectivity to support administrative operations.

Human Resources (HR)

The HR department requires secure access to employee records, internal communication systems, printers, and file servers. Access to HR resources should be restricted to authorized personnel only.

Finance Department

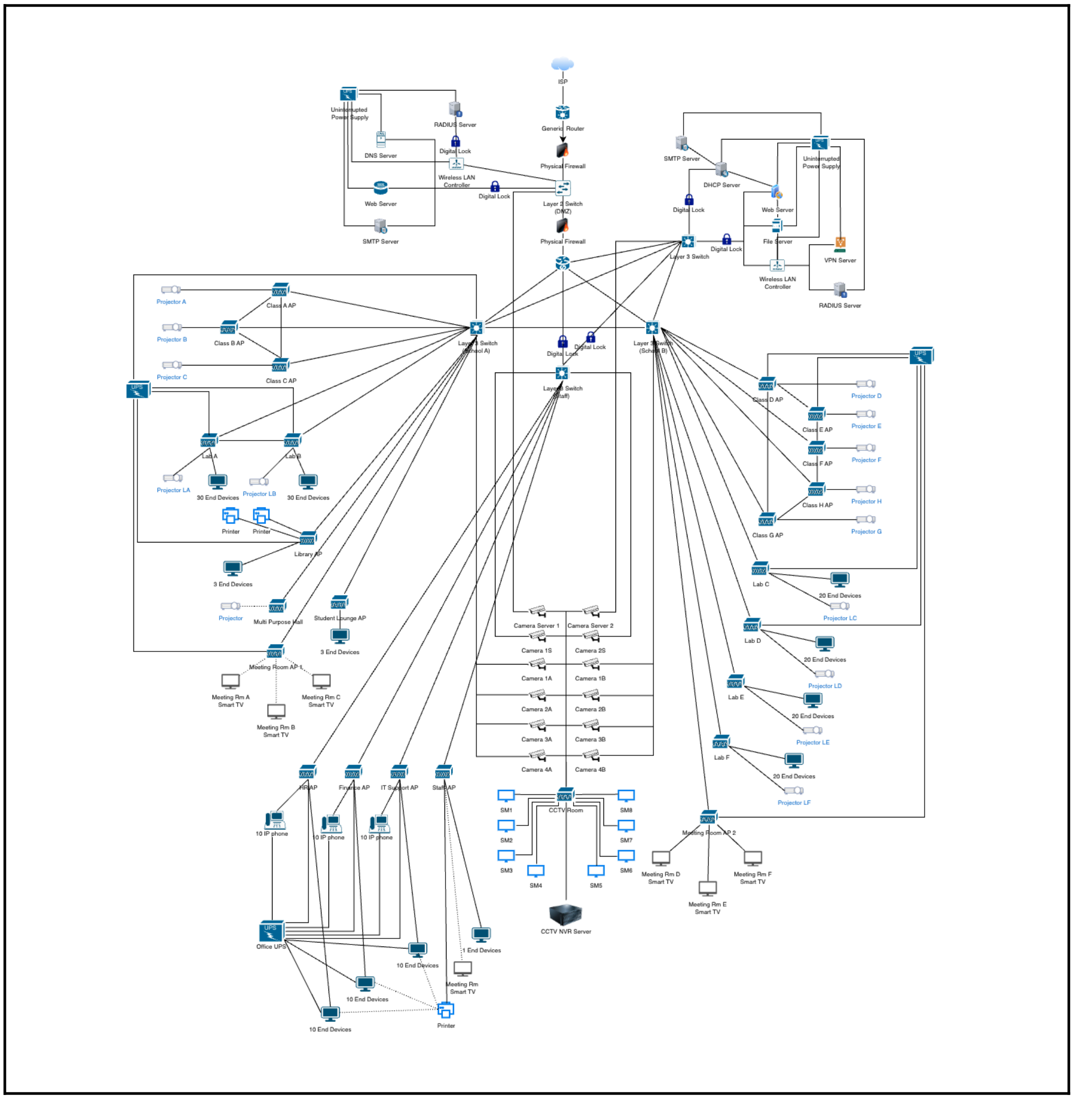
The Finance department handles confidential financial information and requires enhanced network security measures such as VLAN isolation, restricted access policies, and secure server connectivity.

IT Support Department

The IT Support department is responsible for maintaining the campus network infrastructure, servers, and security systems. The department requires administrative access to network devices, monitoring systems, and server management tools.

3. Network Topology Design

3.1. Proposed Network Topology



The proposed campus network uses a hybrid topology combining star, mesh, and bus topologies to provide scalability, reliability, and efficient communication between schools and supporting departments.

3.2. Star-Bus Topology

Star-Bus topology is primarily implemented within School A, School B, laboratories, classrooms, meeting rooms, and supporting departments, where end-user devices connect to centralised switches or wireless access points. By using a hybrid between star and bus topology, it allows for easy management through the star network, while also simplifying potential expansions from the bus network, there is also an added advantage of reducing cabling costs.

Devices connected within the star topology include:

- Desktop PCs and laptops
- Smart TVs and smart displays
- Projectors
- Printers and photocopiers
- IP phones
- IoT devices
- Wireless access points

This topology improves network management by centralising communication through managed switches and access points. It also simplifies troubleshooting and maintenance because failures in individual devices or cables do not affect the entire network. In addition, star topology allows for easier expansion of classrooms, laboratories, and departmental facilities in the future.

3.3. Mesh Topology

Mesh topology is implemented between core Layer 3 switches, routers, firewalls, and server infrastructure to increase redundancy and network reliability. Multiple interconnected backbone paths are used to ensure continuous communication between critical network devices and services.

The mesh backbone helps reduce risks of single point of failures by providing alternative communication paths if a network device or connection path becomes unavailable. This is especially important for maintaining access to centralised services such as:

- DHCP servers
- DNS servers
- Web servers
- File servers

- SMTP servers
- VPN servers
- Wireless LAN controllers
- RADIUS authentication servers

The implementation of mesh connectivity also improves fault tolerance and overall network stability within the campus environment.

3.4. Bus Topology

A backbone bus topology is used to interconnect major network segments across the campus environment. The backbone connection acts as the central communication pathway linking School A, School B, supporting departments, server infrastructure, and Internet services.

The backbone network supports:

- Inter-VLAN communication
- Centralized server access
- Internet connectivity
- Wireless network management
- Communication between departments and schools

High-speed Ethernet connections and managed switching infrastructure are used throughout the backbone network to ensure stable data transmission and efficient traffic handling across the campus.

The hybrid topology was selected because it combines the advantages of multiple topology types while minimising their limitations. Star topology provides simplicity and ease of management for end-user devices, mesh topology improves redundancy and reliability for critical infrastructure, and bus topology enables efficient backbone communication between network segments.

This combination provides:

- Scalability for future expansion
- Improved fault tolerance
- Efficient traffic management
- Better network performance
- Centralized management
- Enhanced security through VLAN segmentation
- Reliable connectivity for smart campus technologies

Overall, the proposed hybrid topology provides a cost-effective and scalable solution suitable for a medium-sized educational campus environment.

3.5. Backbone Connectivity

The campus backbone network is designed to provide high-speed and reliable communication between School A and School B, supporting departments, server infrastructure, and Internet services. The backbone infrastructure interconnects the major network segments through core Layer 3 switches, routers, and managed switching devices to support centralized communication across the campus environment.

The proposed hybrid topology was selected because it combines the strengths of multiple topology types while minimising their limitations. Star topology improves network manageability and scalability within classrooms, laboratories, and departments by centralising device connections through switches and wireless access points. Mesh topology enhances reliability and redundancy between critical backbone devices by providing multiple communication paths and reducing the risk of a single point of failure. Bus topology supports efficient backbone communication by linking major network segments through a centralised backbone infrastructure.

The backbone network also supports:

- Inter-VLAN communication
- Centralised server access
- Wireless network management
- Internet connectivity
- Communication between departments and schools

High-speed Ethernet connections and managed network devices are implemented throughout the backbone infrastructure to ensure stable data transmission, efficient traffic management, and future scalability for the campus network.

Overall, the proposed backbone design provides a scalable, secure, reliable, and cost-effective network solution suitable for a medium-sized educational campus environment.

3.6. LAN Design

The LAN design of the campus primarily uses a star topology, where end-user devices connect to centralised Layer 3 switches through access points and network switches. Separate LAN environments are implemented for classrooms, laboratories, departments, meeting rooms, and administrative areas.

School A and School B each contain dedicated classroom access points, laboratories, projectors, smart displays, printers, and student devices connected through managed switching infrastructure. Supporting departments such as HR, Finance, and IT Support are also connected through separate LAN segments to improve network organisation and security.

VLAN segmentation is implemented to logically separate:

- Student networks
- Staff networks
- HR department
- Finance department
- IT Support department
- Server infrastructure
- Wireless networks

This segmentation reduces broadcast traffic, improves network performance, and prevents unauthorised access between departments.

The LAN infrastructure supports:

- Wired and wireless connectivity
- Smart classroom technologies
- IP telephony services
- Centralized server access
- Internet connectivity
- Multimedia and IoT devices

Managed switches and wireless access points are used throughout the campus to improve scalability, simplify network management, and support future expansion requirements.

4. Network Segmentation (VLAN Design)

Student VLANs	
Students	VLAN 50
Teachers VLANs	
Teachers	VLAN 40

Support Staff VLANs	
IT Support	VLAN 10
Finance	VLAN 20
HR	VLAN 30
Server VLANs	
Server	VLAN 60
CCTV	VLAN 70
Meeting room	VLAN 80
Wireless AP	VLAN 90
Guest WiFi	VLAN 100

5. IPv4 Addressing and Network Subnetting

5.1. IP Addressing Scheme

10.0.0.0/8

5.2. Subnet Allocation

VLAN Name	VLAN	IPv4 Addressing
VLAN 10 — IT Support	10.10.10.0/24	255.255.255.0
VLAN 20 — Finance	10.20.20.0/24	255.255.255.0
VLAN 30 — HR	10.30.30.0/24	255.255.255.0
VLAN 40 — Staff	10.40.40.0/24	255.255.255.0
VLAN 50 — Students(A)	10.50.10.0/22	255.255.255.0
VLAN 50 — Students(B)	10.50.20.0/22	255.255.255.0
VLAN 60 — Servers	10.60.60.0/24	255.255.255.0
VLAN 70 — CCTV	10.70.70.0/24	255.255.255.0
VLAN 80 — Meeting Room	10.80.80.0/24	255.255.255.0
VLAN 90 — Wireless AP	10.90.90.0/24	255.255.255.0
VLAN 100 — Guest WiFi	10.100.100.0/22	255.255.255.0

5.3. Subnetting Calculations

Students VLAN: $2^{(32-22)}-2 = 1022$

Staff VLAN: $2^{(32-24)}-2 = 254$

Guest VLAN: $2^{(32-22)}-2 = 1022$

5.4. Address Allocation Table

Server Allocation:

Device	IP Address
DNS Server	10.60.60.10

DHCP Server	10.60.60.11
Web Server	10.60.60.12
SMTP Server	10.60.60.13
File Server	10.60.60.14
VPN Server	10.60.60.15
RADIUS Server	10.60.60.16
Wireless LAN Controller Server	10.60.60.17

CCTV Allocation:

Device	**IP Address**
CCTV NVR Server	10.70.70.10
Cameras	10.70.70.20 –10.70.70.32

Wireless AP Allocation：

Access Points	**IP Address**
Class APs (School A)	10.90.90.10 —10.90.90.19
Class APs (School B)	10.90.90.20 —10.90.90.30
Lab APs(School A)	10.90.90.31 —10.90.90.40
Lab APs(School B)	10.90.90.41 —10.90.90.50
Meeting Room AP(School A)	10.90.90.51—10.90.90.60
Meeting Room AP(School B)	10.90.90.61—10.90.90.70

IP Phone Allocation:

Department	IP Address
HR	10.30.30.10 —10.30.30.20
Finance	10.20.20.10 —10.20.20.20
IT Support	10.10.10.10 —10.10.10.20

Device:

Device	IP Address
Meeting RM A Smart TV（School A）	10.80.80.10
Meeting RM B Smart TV（School A）	10.80.80.11
Meeting RM C Smart TV（School A）	10.80.80.12

6. Network Devices and Connectivity

The proposed smart campus network infrastructure utilises enterprise-grade networking devices to provide reliable, scalable, and secure communication across School A, School B, and the supporting departments. The selected devices were chosen based on the requirements of the proposed hybrid topology, centralised server infrastructure, VLAN implementation, wireless connectivity, and future scalability within a medium-sized campus environment.

The network design integrates routers, Layer 3 managed switches, wireless access points, firewall appliances, and Cat6 Ethernet cabling to support approximately 300 students, teaching staff, administrative departments, and smart classroom technologies.

6.1. Routers

The proposed topology utilises TP-Link Omada ER8411 enterprise routers to manage communication between the campus backbone infrastructure, VLAN environments, VPN services, and external Internet connectivity.

The routers are implemented at the core of the network infrastructure to:

- Route traffic between network segments
- Provide Internet gateway access
- Support VPN communication
- Manage communication between schools and supporting departments
- Connect the campus backbone to external networks

The ER8411 routers were selected because they support high-speed enterprise networking features such as:

- Multi-gigabit throughput
- VPN functionality
- Load balancing
- Centralised network management
- Advanced routing capabilities

These features are important in the proposed smart campus environment because the network supports large volumes of simultaneous traffic from classrooms, laboratories, wireless devices, smart displays, IP phones, and centralised server systems.

The implementation of centralised routing infrastructure improves traffic management, network scalability, and communication efficiency between School A, School B, and the supporting departments.

6.2. Switches

The proposed network design utilises both core Layer 3 switches and managed PoE access switches to support communication across the campus infrastructure.

The backbone infrastructure implements Ubiquiti UniFi Pro Aggregation Layer 3 switches to manage high-speed backbone communication between:

- Schools
- Supporting departments
- Server infrastructure
- Wireless controllers
- Firewall appliances

Layer 3 switching was selected because it supports:

- Inter-VLAN communication
- Traffic segmentation
- Faster internal routing
- Improved scalability
- Better traffic management

The use of Layer 3 switches is particularly important within the proposed topology because the campus network is segmented into multiple VLANs, including student, staff, department, and server VLANs.

The campus also utilises Ubiquiti UniFi USW-48-POE managed PoE switches within classrooms, laboratories, and department offices. These switches provide centralised connectivity for:

- Desktop PCs
- IP phones
- Wireless access points
- Smart classroom devices
- Printers

PoE functionality allows devices such as wireless access points and IP phones to receive both power and network connectivity through a single Ethernet cable, reducing cabling complexity and simplifying deployment.

Managed switching infrastructure improves network organisation, centralised administration, troubleshooting efficiency, and future scalability throughout the campus environment.

6.3. Wireless Access Points

The proposed smart campus environment implements Ubiquiti UniFi U7 Lite wireless access points throughout classrooms, laboratories, meeting rooms, and supporting department offices to provide campus-wide wireless connectivity.

The wireless infrastructure supports:

- Student laptops and mobile devices
- Teacher laptops
- Smart classroom technologies
- IoT devices
- Wireless Internet access

Wireless access point was selected as the proposed campus environment relies heavily on mobile connectivity and digital learning systems. The implementation of centralised wireless infrastructure improves user mobility by allowing students and staff to access campus resources from different locations without requiring wired connections. It replaces the Hub, which does not provide any security and is less advanced.

To improve wireless management and security, the topology also integrates Cisco Catalyst 9800-L wireless LAN controllers. Centralised wireless management allows IT administrators to:

- Monitor wireless traffic
- Configure access points centrally
- Manage wireless security policies
- Improve network performance
- Control user authentication

This implementation improves scalability and simplifies wireless administration across the campus environment.

6.4. Ethernet Cabling (Cat6)

The proposed network infrastructure utilises Cat6 Ethernet cabling to provide high-speed wired connectivity between routers, switches, servers, wireless access points, desktop workstations, and smart classroom devices.

Cat6 cabling was selected because it supports:

- Gigabit Ethernet communication
- Higher bandwidth capacity
- Reduced signal interference
- Improved transmission reliability
- Better future scalability

The use of structured Cat6 cabling is particularly important within the proposed smart campus environment because the network supports bandwidth-intensive services such as:

- Smart classroom systems
- Multimedia learning platforms
- VoIP communication
- File sharing
- Centralized server access
- Wireless backhaul communication

Structured cabling also improves network organisation and simplifies future maintenance, troubleshooting, and infrastructure expansion.

6.5. Internet Connectivity

The proposed campus network implements centralised Internet connectivity to support online learning platforms, cloud services, web applications, email communication, VPN services, and administrative operations.

Internet access is distributed throughout School A, School B, and supporting departments through the backbone infrastructure and Layer 3 switching environment.

The topology integrates Fortinet FortiGate 60F firewall appliances between the campus network and the Internet gateway to improve security and protect internal network resources from unauthorised external access.

The firewall infrastructure supports:

- Traffic filtering
- VPN security
- Access control policies
- Threat monitoring
- Network protection

The centralised Internet infrastructure is necessary because the proposed smart campus environment relies heavily on cloud-based educational platforms, online communication systems, and wireless connectivity for daily academic and administrative operations.

The implementation of centralised Internet connectivity improves communication efficiency, accessibility to digital resources, and overall network performance across the campus environment.

7. Hardware and Equipment List (Costing)

The selected hardware and network equipment were chosen based on the scalability, reliability, performance, compatibility, and cost-effectiveness, suitable for a medium-sized campus environment. Enterprise-grade devices such as Layer 3 switches, managed access points, and centralised server systems were selected to support stable communication, VLAN implementation, wireless connectivity, and future expansion requirements.

7.1. Desktop PCs

Equipment	Brand / Model	Quantity	Unit Price (SGD)	Total Cost (SGD)
Student Workstations	Dell OptiPlex 7020 Desktop	160	1,200	192,000
Teacher Workstations	HP Pro Mini 400 G9	20	1,500	30,000
HR Department PCs	HP Pro Mini 400 G9	10	1,500	15,000
Finance Department PCs	HP Pro Mini 400 G9	10	1,500	15,000
IT Support Workstations	Lenovo ThinkCentre Neo 50t	10	1,800	18,000
Monitors	Dell 24 Monitor P2425H	210	250	52,500
Subtotal				322,500

7.2. Laptops

Equipment	Brand / Model	Quantity	Unit Price (SGD)	Total Cost (SGD)
Teacher Laptops	Lenovo ThinkPad E14 Gen 5	20	1,800	36,000
Subtotal				36,000

7.3. Servers

Equipment	Brand/Model	Quantity	Unit Price (SGD)	Total Cost (SGD)
DHCP, DNS, SMTP, Web and File **Servers**	Dell PowerEdge T360	6	6,000	36,000
VPN Server	Dell PowerEdge T360	1	6,000	6,000
RADIUS Authentication Servers	Dell PowerEdge T360	2	6,000	12,000
Backup Storage NAS	Synology DiskStation DS1821+	2	2,000	4,000
UPS System	APC Smart-UPS	8	1,200	9,600
Server Rack Cabinet	APC NetShelter Rack	4	2,000	8,000
CCTV NVR Server	Hikvision DS-7732NI-M4	1	2500	2500
Subtotal				78,100

7.4. IP Phones

Equipment	Brand/Model	Quantity	Unit Price (SGD)	Total Cost (SGD)
IP Phones	Yealink SIP-T31P	60	120	7,200
Subtotal				7,200

7.5. Smart TVs and Smart Displays

Equipment	Brand/Model	Quantity	Unit Price (SGD)	Total Cost (SGD)
Smart Classroom Displays	Samsung Flip Pro	20	4,000	80,000
Multimedia Projectors	Epson EB-FH52	15	1,200	18,000
Subtotal				98,000

7.6. Printers and Photocopying Machines

Equipment	Brand/Model	Quantity	Unit Price (SGD)	Total Cost (SGD)
Network Printers	Canon MAXIFY GX7070	15	800	12,000
Photocopying Machines	FujiFilm Apeos C3060	4	3500	14,000
Subtotal				26,000

7.7. Network Equipment Costing

Equipment	Brand/Model	Quantity	Unit Price (SGD)	Total Cost (SGD)
Core Layer 3 Switches	Ubiquiti UniFi Pro Aggregation	4	1,500	6,000
Managed Access	Ubiquiti UniFi	20	1,000	20,000

Switches	USW-48-POE			
Core Routers	TP-Link Omada ER8411	4	900	3,600
Wireless Access Points	Ubiquiti UniFi U7 Lite	30	250	7,500
Wireless LAN Controllers	Cisco Catalyst 9800-L	2	3,000	6,000
Firewall Appliances	ForTinet FortiGate 60F	2	3,500	7,000
Digital Door Locks	Samsung SHP-DP609 Digital Door Lock	23	450	10,350
CCTV IP Cameras	Hikvision DS-2CD2143G2-I	12	250	3000
Microsoft 365 Campus Licensing	Microsoft 365 Business	1	35,000	35,000
Endpoint Antivirus Protections	Bitdefender GravityZone Business	1	10,000	10,000
Cloud Backup Subscription	Synology Cloud Backup Services	1	4,000	4,000
VPN & Remote Access Licensing	Fortinet Secure VPN Licensing	1	5,000	5,000
Miscellaneous Network Accessories	Patch Panels, Connectors, Adapters	1	10,000	10,000
Singtel Internet Service	Enterprise Fibre (Static IP) 1 - 10 Gbps Plan	1	1,000	1,000
MyRepublic Internet Service	Ultra 10Gbps Fibre	1	1,000	1,000
Subtotal				132,450

The proposed equipment provides sufficient performance and scalability to support approximately 300 students, teaching staff, and administrative departments while remaining within the allocated project budget.

8. Server Design

The proposed campus network implements a centralised server infrastructure to support communication, network management, resource sharing, security, and academic services across School A, School B, and the supporting departments. Centralized servers were selected because they improve scalability, simplify network administration, reduce management complexity, and provide more efficient control of campus-wide services.

The server infrastructure supports approximately 300 students, teaching staff, and administrative personnel while ensuring reliable access to learning resources, Internet services, and internal campus systems. Centralising the servers also improves security and maintenance by allowing IT Support staff to manage critical services from a single server environment.

8.1. DHCP Server

The Dynamic Host Configuration Protocol (DHCP) server is implemented to automatically assign IP addresses and network configuration settings to devices connected to the campus network. The use of a DHCP server reduces the need for manual IP configuration and minimises configuration errors within the campus environment.

The DHCP server automatically distributes:

- IP addresses
- Subnet masks
- Default gateway addresses
- DNS server information

This service is especially important because the campus network contains a large number of devices, including:

- Student workstations
- Teacher computers
- Department PCs

- Wireless devices
- IP phones
- Smart classroom devices

Without DHCP, manually configuring hundreds of devices would increase administrative workload and the likelihood of IP conflicts. The implementation of DHCP improves scalability, simplifies network management, and allows devices to connect to the network more efficiently.

8.2. DNS Server

The Domain Name System (DNS) server is implemented to translate domain names into IP addresses within the campus network. DNS improves usability because users can access servers and network services using user-friendly domain names instead of numerical IP addresses.

The DNS server supports communication between:

- Web servers
- File servers
- Email servers
- Wireless services
- Internet resources

The implementation of a centralized DNS server improves accessibility and communication efficiency across the campus network. It also simplifies network administration because changes to server addresses can be centrally managed without requiring reconfiguration on every client device.

DNS is particularly important within the proposed smart campus environment because students and staff frequently access online learning systems, web portals, and shared network resources.

8.3. Web Server

The web server is implemented to host campus-related websites and web-based applications. These services may include:

- Student learning portals
- Staff portals
- School websites
- Internal campus systems

- Academic resources

The implementation of a centralised web server improves accessibility to educational materials and online services for both students and staff. Since the campus environment supports smart classrooms and wireless connectivity, a web server is necessary to provide centralised access to learning content and online communication platforms.

The web server also supports future scalability by allowing additional web applications and digital learning services to be integrated into the campus network infrastructure.

8.4. File Server

The file server is implemented to provide centralised storage and file-sharing services throughout the campus environment. The file server allows students, teachers, HR staff, Finance personnel, and IT Support staff to securely access shared resources and documents through the network.

The file server supports:

- Centralised data storage
- Teaching material distribution
- Administrative document sharing
- Backup and recovery operations
- Controlled access permissions

A centralised file server improves data organisation and reduces the risk of data loss caused by storing files individually on local devices. It also improves collaboration between departments and teaching staff while allowing IT administrators to manage permissions and backups more efficiently.

Access control mechanisms are implemented to ensure sensitive files from departments such as HR and Finance remain accessible only to authorised personnel.

8.5. Email (SMTP) Server

The Simple Mail Transfer Protocol (SMTP) server is implemented to support email communication across the campus environment. Email services are necessary to facilitate communication between students, teachers, administrative departments, and IT Support staff.

The SMTP server supports:

- Internal communication between departments
- Academic announcements
- Student notifications
- Administrative communication
- File and information sharing

The implementation of a centralised email server improves communication efficiency and ensures official campus communication can be managed securely within the network infrastructure.

Email communication is especially important in the proposed smart campus environment because many educational and administrative processes rely on digital communication and online information sharing. The centralised email system also allows IT administrators to manage user accounts, improve security policies, and monitor communication services more effectively.

8.6. VPN Server

The Virtual Private Network (VPN) server is implemented within the proposed campus network to provide secure remote access to internal network resources for authorised staff and administrators. The inclusion of a VPN server in the network design improves flexibility by allowing users to securely connect to the campus infrastructure from external locations through encrypted Internet connections.

The VPN server supports secure access to:

- File servers
- Email services
- Administrative systems
- Internal campus resources
- Network management systems

The implementation of a VPN server is justified because the proposed smart campus environment relies heavily on centralised digital services and online communication. Teaching staff, administrative personnel, and IT Support staff may require remote access to academic resources, files, or management systems outside the campus environment.

Without a VPN solution, sensitive campus data could be exposed to security risks when accessed through unsecured Internet connections. The VPN server improves security by encrypting transmitted data and reducing the risk of unauthorised access, interception, or data breaches.

The VPN server also supports the scalability of the proposed network design by allowing future remote learning, hybrid working arrangements, and off-campus administration to be integrated into the campus infrastructure more efficiently.

8.7. RADIUS Authentication Server

The Remote Authentication Dial-In User Service (RADIUS) server is implemented to provide centralised authentication, authorisation, and access control services across the campus network infrastructure. The inclusion of a RADIUS server improves network security by ensuring only authorised users are allowed to access restricted network resources and services.

The RADIUS server is integrated with:

- Wireless access points
- VPN services
- Administrative systems
- Network devices
- Staff authentication systems

The implementation of centralised authentication is necessary because the proposed campus network contains multiple user groups, including students, teachers, HR staff, Finance personnel, and IT administrators. Each group requires different levels of access to network resources and services.

The RADIUS server improves the overall security of the proposed solution by supporting:

- User authentication
- Access control policies
- Activity logging and monitoring
- Centralised credential management

The implementation of a RADIUS server also complements the VLAN segmentation and firewall security measures used within the proposed topology. By combining centralised authentication with VLAN separation, the network design provides stronger protection against unauthorised access and improves management efficiency for the IT Support department.

Overall, the inclusion of a RADIUS authentication server strengthens the reliability, security, and manageability of the smart campus network infrastructure.

9. Security Policies

The school network applied layered security to safeguard resources, regulate access, and ensure reliable connectivity. Policies use VLAN isolation, firewalls, and authentication to establish clear boundaries for students, employees, and servers. These safeguards maintain safe operations in academic and administrative settings, protect sensitive data, and ensure responsible usage.

9.1. Staff and Student Access Control

Students use classroom and laboratory access points to connect via the Student VLAN, where their permissions are restricted to learning platforms, online educational resources, and general Internet access. This limitation reduces the risk of students accessing confidential organisational systems, administrative databases, and server infrastructure located within the Staff and Server VLANs. The separation of student traffic also improves network performance and helps reduce unnecessary congestion across administrative network segments.

Employees connect through the Staff VLAN to access internal file servers, printers, email systems, Microsoft 365 campus services, and administrative platforms required for daily academic and operational activities. Staff members are provided with higher access privileges compared to students because teaching staff and supporting departments require access to centralized servers, cloud-based services, and departmental resources within the campus environment.

The RADIUS server and Wireless LAN Controller manage authentication for both groups, ensuring that only registered users and authorized devices can connect to the campus network infrastructure. Centralized authentication also allows the IT department to monitor user access, manage permissions, and improve overall network security across School A, School B, and the supporting departments.

Unauthorised access to wired and Wi-Fi connections is prevented through additional security measures such as MAC address filtering, user credentials, endpoint antivirus protection, and VPN authentication controls. These security measures help protect the campus infrastructure from unauthorized devices, malware infections, and potential cyber threats.

Guest Wi-Fi is kept separate from internal VLANs to further protect sensitive resources and prevent visitors from interfacing with staff or student network activity. This separation improves security by isolating external users from centralized servers, administrative systems, and internal campus communications while still providing controlled Internet connectivity for guests and visitors.

9.2. Firewall and Network Security

The Physical firewall positioned between router and the Layer 3 switches plays a critical role in filtering both inbound and outbound traffic. This firewall acts as the first line of defense, inspecting all packets entering or leaving the network and blocking any unauthorised or suspicious connections. In order to protect internal VLANS from outside threats, public-facing servers like DNS, Web and SMTP are housed within the DMS segment behind the firewall. Additional protections such as intrusion prevention, antivirus scanning and detailed logging are enabled to detect and block suspicious activity. By adopting these safeguards, the firewall creates a distinct division between the internal networks of School A and School B and external internet traffic, protecting sensitive resources while allowing needed communication.

9.3. VLAN Security

VLAN segmentation is used to isolate data between students, staff, departments, loT devices, and servers, ensuring that each group operates within its own safe area. Access control lists on the Layer 3 switches carefully regulate inter-VLAN connectivity, preventing students from accessing staff or server VLANs while enabling staff to access shared resources without interfering with student devices. Unauthorised segment hopping is prevented by configuring trunk ports with VLAN tagging, which authorised VLANs are permitted. DHCP snooping is enabled to validate IP address assignments, ensuring that rogue DHCP servers cannot disrupt the network. By blocking IP spoofing attacks and preserving the security of device connections, dynamic ARP inspection adds an additional line of defence. VLAN segmentation is extended to VPN connections for remote access, ensuring that users connecting from outside the campus are placed in secure, separated VLANs with limited permissions. This stops unauthorized external users from directly accessing internal resources.

9.4. Password and Authentication Policies

Strong password and authentication policies are enforced across the school network to protect sensitive resources. All users are required to create passwords with a minimum of eight characters, including uppercase and lowercase letters, numbers, and symbols. Passwords cannot be reused and expire after ninety days in order to maintain security. Multi-factor authentication is required for server and switch administrative accounts in order to prevent unwelcome entry. While VPN users must first authenticate in order to obtain safe remote access to the network, centralised authentication via the RADIUS Server guarantees constant enforcement of these rules across wired and wireless connections.

9.5. Internet and Email Access Policies

Internet and email usage within the school network is carefully regulated to maintain security and productivity. Employees are allowed to browse for work related purposes, but students are restricted from accessing non-academic websites like social networking and gaming platforms. The SMTP server scans outgoing emails for viruses and spam, and the firewall blocks hazardous or inappropriate information by combining web filtering and SSL inspection. In addition to being checked for viruses, email attachments are also checked for file type limits. Only authorised employees are able to access VPNs for remote work, which enables safe, secure communication with the school's internal systems.

10. Facilities Layout

10.1. Computer Labs

The Student VLAN connects the computer laboratories, enabling safe access to online resources and e-learning platforms. Every lab switch is directly connected to the Layer 3 switch, ensuring consistent performance and fast connectivity. While lab printers and projectors function under the same VLAN for convenience, devices are set up for cable connections to ensure stability. Each lab is equipped with thirty end devices, allowing students to perform research, coding, and practical exercises effectively. In order to keep student activities separate from staff operations, network restriction limits access to the administrative system. The firewall and authentication servers keep an eye on traffic to stop unwanted access.

10.2. Classrooms

Classrooms are equipped with smart boards, projectors, and wireless access points that link to the loT VLAN are all installed in classrooms. Teachers can safely access internal servers and teaching materials by using the Staff VLAN on their computers and tablets. All classroom access points are managed by the Wireless LAN Controller, which also ensures consistent bandwidth control and authentication throughout the network. With strong Wi-fi coverage controlled by the Wireless LAN Controller to provide uniform bandwidth and authentication across all devices, every classroom is built to allow interactive learning. The network design allows lag-free online collaboration, video streaming, and multimedia education.

10.3. Staff Offices

The Staff VLAN is used by staff offices to facilitate secure communication and administrative tasks. VoIP phones, office computers, and printers are connected to the Layer 3 backbone via managed switches. Firewalls and authentication procedures safeguard access to internal computers and databases, ensuring the privacy of important data. High speed wired connections are installed in every business to facilitate routine tasks including internal communication, financial administration, and document processing. Additionally, the network contains centralised file sharing and printing features, ensuring smooth staff communication while maintaining strong access control.

10.4. Meeting Rooms

loT gadgets like smart TVs and conferencing systems are combined with staff computers linked to the Staff VLAN in meeting rooms. Prioritising bandwidth guarantees seamless streaming of presentations and video conferences. Each conference room's wireless access points are set up to divide staff and visitor traffic, preserving security while increasing cooperation. These access points are handled by the Wireless LAN Controller, which enables reliable authentication and performance monitoring. Important data shared during meetings is also kept safe by the firewall and VLAN segmentation, which protects internal resources from external devices.

10.5. Supporting Departments

The Department VLAN is used for supporting departments including IT, HR, and finance. To stop illegal data exchange, each department has limited inter-VLAN connectivity. In order to maintain the network's security and operational effectiveness throughout the entire campus, the IT department controls servers, firewalls, and monitoring tools situated in the Server VLAN. The Finance department handles budget and transactions using encrypted communication channels while the HR department manages staff records and payroll systems through secure access to internal servers.

11. Budget Breakdown (Analysis)

The proposed smart campus network infrastructure has an estimated total cost of SGD 645,750, which remains within the allocated project budget of SGD 800,000. The budget was distributed across networking infrastructure, end-user devices, server systems, and security technologies to ensure the campus environment supports reliable communication, centralised management, scalability, and security.

A large portion of the budget was allocated to end-user devices and smart classroom technologies because the campus environment requires a significant number of desktop workstations, laptops, smart displays, and multimedia systems to support approximately 300 students and teaching staff members. The implementation of smart classroom technologies was prioritised to improve digital learning, classroom interactivity, and accessibility to online educational resources.

Enterprise-grade networking equipment, such as Layer 3 switches, managed PoE switches, wireless access points, and centralised routing infrastructure, was selected to support the proposed hybrid topology and VLAN-based network segmentation. Although these devices increase the initial implementation cost, they provide better scalability, centralised management, improved security, and long-term reliability compared to lower-cost consumer-grade alternatives.

The server infrastructure was also allocated a significant portion of the budget because the proposed campus environment relies heavily on centralised services such as DHCP, DNS, VPN, RADIUS authentication, file sharing, email communication, and web-based learning systems. Centralized servers simplify administration and improve communication between School A, School B, and the supporting departments.

Security systems, including firewall appliances, VPN services, endpoint antivirus protection, digital door locks, CCTV infrastructure, and centralised authentication systems, were implemented to strengthen both cybersecurity and physical security within the campus environment. These systems help protect the campus network from cyber threats, malware infections, unauthorised access, and potential data breaches. The security infrastructure is particularly important because departments such as HR and Finance manage confidential organisational information that requires restricted access, secure communication, and continuous protection of sensitive data.

Although the proposed infrastructure already supports the current campus requirements, approximately SGD 99,750 remains from the allocated project budget. Rather than fully exhausting the budget during the initial implementation phase, the remaining funds may be reserved for future expansion, maintenance, contingency planning, and infrastructure upgrades.

Future scalability is an important consideration within the proposed smart campus design. As student enrollment increases, additional classrooms, laboratories, wireless access points, and workstation systems may be required to support larger numbers of concurrent users and connected devices. The implementation of enterprise-grade Layer 3 switches and centralised wireless infrastructure allows the network to scale more efficiently without requiring a complete redesign of the topology.

The remaining budget may also support future upgrades, such as:

- Additional computer laboratories
- Expansion of wireless coverage
- Higher-capacity servers and storage systems
- Cloud-based learning platforms
- Advanced cybersecurity solutions
- Biometric access control systems
- Smart IoT classroom technologies
- AI-assisted campus monitoring systems
- Backup Internet connections for redundancy

In the future, the campus may also transition towards hybrid learning environments where students and staff access learning resources remotely through cloud services and VPN connectivity. Because the proposed solution already includes centralised authentication, VPN services, and scalable server infrastructure, the network can support future digital transformation requirements more effectively.

Overall, the proposed budget allocation balances current operational requirements with long-term scalability, allowing the campus network infrastructure to remain reliable, secure, and adaptable for future technological developments.

Budget Category	Total Cost (SGD)
Networking Equipment	56,100
End-User Devices	489,700
Server Infrastructure	117,100
Security Systems	37,350
Grand Total	700,250

12. Conclusion

In conclusion, the proposed smart campus network infrastructure successfully addresses the operational, academic, and security requirements of School A, School B, and the supporting departments. The network design was developed to provide reliable communication, centralised management, secure access control, and scalable connectivity while remaining within the allocated project budget.

The implementation of a hybrid topology combining star, mesh, and backbone connectivity allows the campus network to achieve improved scalability, redundancy, and efficient communication between different network segments. The use of enterprise-grade routers, Layer 3 switches, managed wireless infrastructure, and centralised server systems ensures the network can support approximately 300 students, teaching staff, and administrative personnel within a stable and high-performance environment.

The proposed solution also integrates VLAN segmentation, centralised authentication, VPN connectivity, firewall protection, and endpoint security systems to strengthen both cybersecurity and physical security throughout the campus infrastructure. These security measures are particularly important for protecting confidential information managed by departments such as HR and Finance while ensuring secure access to campus resources and online services.

In addition, the implementation of smart classroom technologies, wireless connectivity, centralised file sharing, and web-based services supports digital learning and improves communication efficiency across the campus environment. The proposed server infrastructure also simplifies network administration and provides a foundation for future expansion and technological upgrades.

Although the current infrastructure already satisfies the operational requirements of the campus, the remaining project budget provides flexibility for future scalability, including additional classrooms, expanded wireless coverage, upgraded server systems, cloud-based learning platforms, and advanced cybersecurity technologies.

Overall, the proposed smart campus network design provides a cost-effective, scalable, secure, and future-ready solution capable of supporting both current and future educational and administrative needs within the campus environment.

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